CS-471: Machine Learning

Week 5

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- Practical Aspects of Machine Learning

Recap of Previous Lecture

- What is multinomial linear regression used for?
- How is the weight matrix for multinomial linear regression defined? What is the order of this matrix?
- How is the input matrix for multinomial linear regression defined? What is the order of this matrix?
- How is score matrix calculated in MLR? Order?
- How is probability matrix calculated in MLR? Order?
- What is the loss function used in MLR?
- What is the gradient of this loss function? Update Rule?

Recap of Previous Lecture

- What is a confusion matrix?
- What is accuracy? Precision? Recall? F1-Score? Specificity?
- How is an ROC curve plotted? What is it used for?
- What is AUC? What is it used for?
- What is underfitting? Overfitting?
- How do we determine if a model is underfit? Overfit?
- What is bias? What is variance?
- What is bias-variance tradeoff?

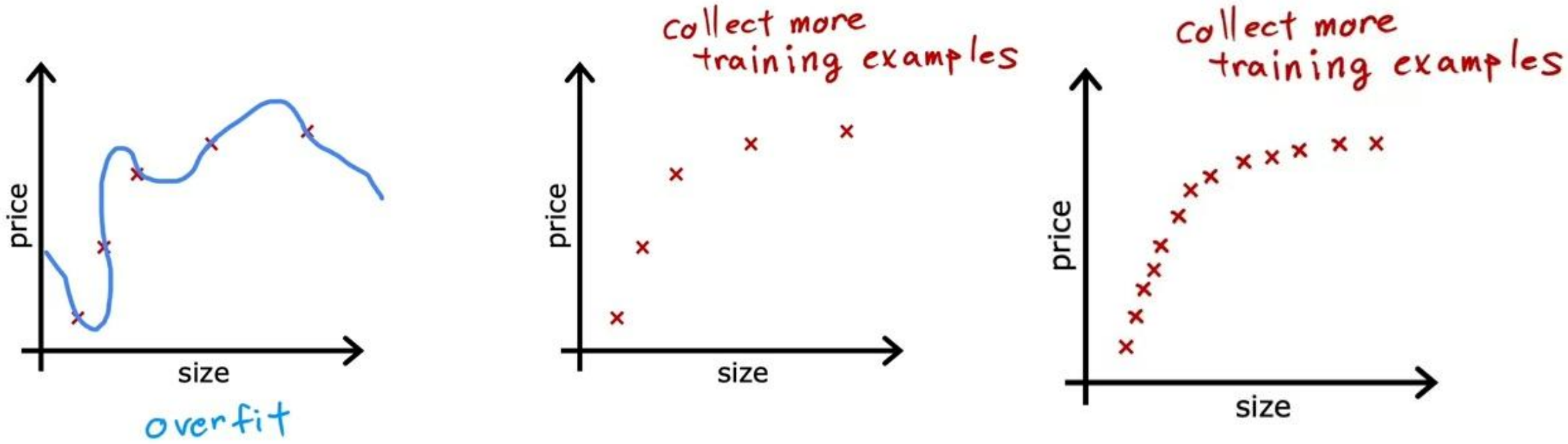
How to prevent Overfitting

- Reduce the complexity of the model
- Use regularization techniques
- Use cross validation
- Early stopping
- Use more Data or do data augmentation

How to prevent Underfitting

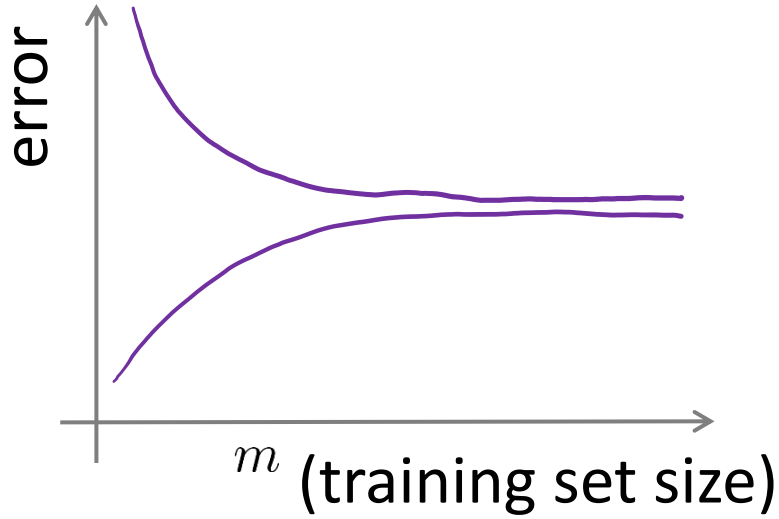
- Increase model complexity
- Train for a longer period of time
- Feature Engineering
- Reduce Regularization
- Try a different machine learning technique

Using more data

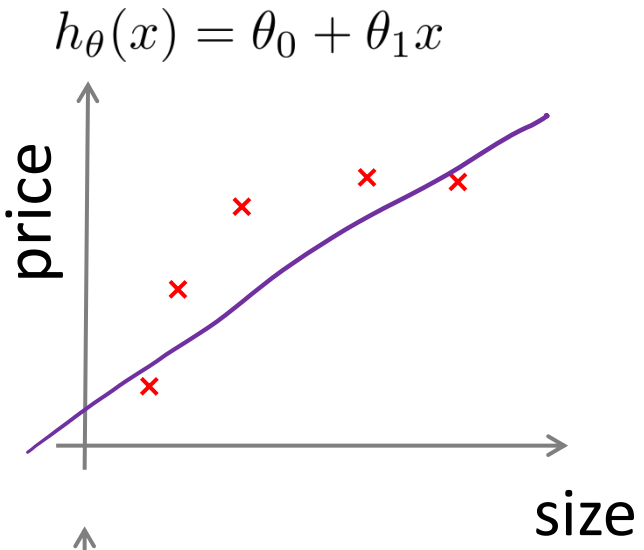


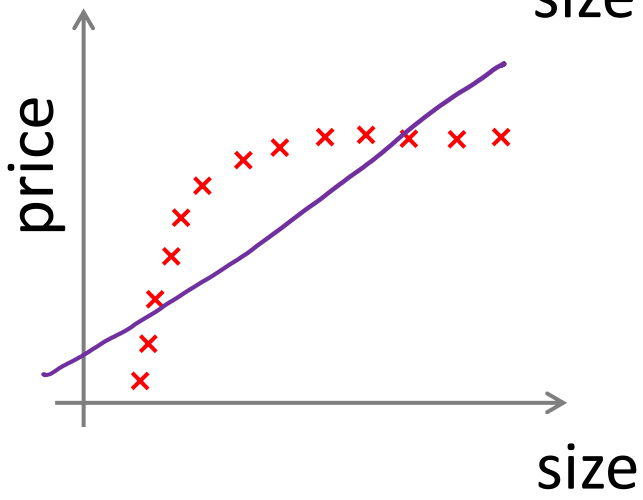
Using more data

High bias



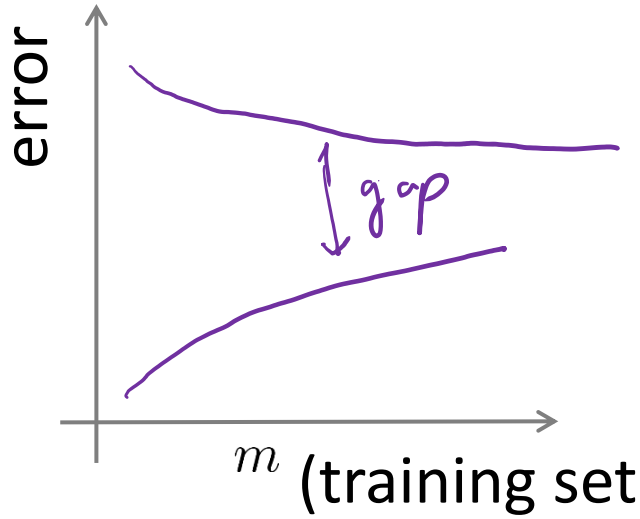
If a learning algorithm is suffering from high bias, getting more training data will not (by itself) help much.





Using more data

High variance



If a learning algorithm is suffering from high variance, getting more training data is likely to help.

$$h_\theta(x) = \theta_0 + \theta_1 x + \cdots + \theta_{100} x^{100}$$

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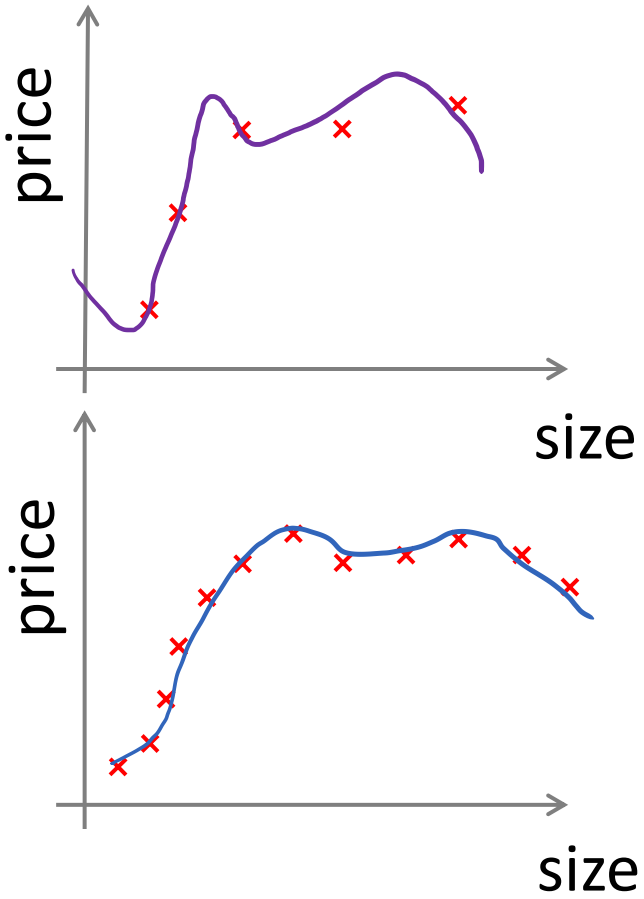

Cross-Validation

- Vanilla cross validation
- K fold cross validation for small datasets
- Leave one out cross validation when very small dataset
- Can be used to avoid overfitting

Questions?